

Reaching the Actionable Radius: Physics-Guided Calibrated Localization of Methane Sources under Wind Uncertainty

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Abstract—Calibrated machine-learning localizers promise guaranteed search regions for methane super-emitters, but a coverage guarantee says nothing about usefulness. On a methane testbed whose closed-form physics makes the *exact* Bayesian posterior computable, we audit what calibration is worth: a conformal-wrapped deep localizer with a verified 90% guarantee delivers a 94–98 m median search radius where the data support 6–7 m. Controlled elimination traces the gap to a single missing classical computation — per-cell matched filtering — and supplying it as *input channels* to a 0.2%-parameter zero-initialized head brings three architecture families into a 1.5 m band at 46–48 m, at the 50 m scale used for facility association in EPA super-emitter notifications. We then make the setting honest: the wind is a *measurement* ($10^\circ/10\%$ per-step error), never the truth. Exact plug-in inversion retains its guarantee only by returning essentially the entire site, while one unified recipe — physics maps parameterized by the declared wind-error budget, provably reducing to the classical matched filter at zero error — degrades gracefully: 82–84 m ignoring the budget, 74–76 m with evidence moments, and 70–73 m with a wind-marginalized goodness-of-fit channel, at unchanged coverage and millisecond inference. Accuracy under wind uncertainty is set by how much of the marginalization the inputs carry: show the network what the physics computes — and what the physics is unsure of.

Index Terms—methane monitoring, conformal prediction, simulation-based inference, wind uncertainty, physics-guided machine learning

I. INTRODUCTION

Methane delivers roughly $80\times$ the 20-year warming potency of CO_2 and decays within about a decade, so rapid repair of oil-and-gas *super-emitters* is among the highest-leverage near-term climate actions. Detection has advanced rapidly — UNEP’s satellite alert system issued over a thousand notifications in its first two years [1] — and attribution is the downstream bottleneck between detection and operator investigation. Regulation supplies a concrete scale: EPA super-emitter notifications associate alerts with facilities at a 50 m radius [2] (program implementation currently extended to 2027), with analogous duties in the EU [3]. We adopt 50 m as an operationally motivated, facility-scale localization target — a region that triggers a focused investigation, not yet a repair.

Continuous monitoring with sparse fixed sensor networks is where such localization can happen: a handful of cheap masts,

always on, catching the plume under naturally meandering wind. Machine learning supplies millisecond inference, and split conformal prediction [4] supplies a guarantee: a region provably containing the source 90% of the time. **The gap is twofold.** First, coverage guarantees frequency, not usefulness: no standard diagnostic [5] can measure how far a calibrated region sits from the smallest region the data support. Second, the classical alternative — exact inversion of the dispersion physics — assumes the wind is *known*, while every real deployment has only a wind *measurement*, and behavior under that mismatch was largely uncharacterized.

Contributions.

- **C1 — An exact-posterior audit.** A methane testbed (CH4-T) whose closed-form physics makes the true posterior of every scenario computable, so conformal *efficiency* can be measured in meters against the information floor (6.2 m at matched coverage). Finding: a verified-90% deep localizer delivers 94–98 m — calibrated is not actionable.
- **C2 — A diagnosis by elimination.** Controlled probes — three encoder families, a strengthened flow-based neural posterior estimator, and the conformalized classical statistic alone — convict a single missing computation, per-cell matched filtering, best supplied as *input channels*: two closed-form maps and a zero-initialized 0.2%-parameter head bring all encoders to 46–48 m with ordinary label training.
- **C3 — One budget-aware recipe that survives measured wind.** The same maps, widened to carry a declared wind-error budget ε (evidence moments and a wind-marginalized misfit channel over the error model), provably reduce to the classical matched filter at $\varepsilon=0$ and remain calibrated at 70–73 m under a realistic $10^\circ/10\%$ budget — where exact inversion built on the same physics returns essentially the entire site.

The components are deliberately classical — matched filters are 1940s detection theory; conformal prediction and label training are standard — and the composition is minimal and fully ablated (Appendix). The exact posterior’s role throughout

Figure 1 placeholder: wind-swept plume, masts, and true source on the 500 m site, with median 90% regions drawn to scale — exact-posterior reference (~ 7 m), our recipe at $\varepsilon=0$ (46–48 m), labels-only baseline (94 m), our recipe under measured wind (70–73 m), and the 50 m facility-association radius.

Fig. 1. The opportunity, to scale (placeholder).

is measurement, not supervision.

II. RELATED WORK

Conformal prediction provides distribution-free coverage [4]; efficiency is the open half of the problem, and we contribute an audit that scores it against the exact Bayesian reference at matched, verified coverage. **Simulation-based inference** amortizes posteriors with neural density estimators [6], [7] at budgets matching ours [8]; its diagnostics test calibration [5], and a growing literature studies robustness when the simulator’s nuisance parameters are misspecified at deployment [9] — our measured-wind design is exactly that setting, addressed by building the declared error model into the inputs. Our maps are *hand-derived sufficient statistics* supplied as spatial channels, a deliberate, auditable counterpoint to learned summaries. **Source-term estimation** inverts dispersion models via optimization or MCMC at minutes-to-hours per event [10]; deep localizers give millisecond inference but lacked instruments certifying how much information they extract — or how they fail when the wind is mismeasured. **Controlled-release evaluation** (METEC) scores continuous-monitoring systems on detection, quantification, and equipment-scale localization [11]; CH4-T mirrors that task structure at super-emitter rates with exact ground truth, complementing — not replacing — field validation.

III. METHODOLOGY

A. Testbed, measured wind, and the audit (C1)

CH4-T places a source of unknown location and rate ($10\text{--}500\text{ kg h}^{-1}$, log-uniform) on a 500×500 m site sensed by 4–12 masts (ppm readings, 0.2–3 ppm noise, dropout). The plume is a steady Gaussian under Pasquill–Gifford stability with Briggs open-country dispersion (coefficients verified against published tables; mass flux conserved to 4×10^{-6}), driven quasi-statically by a fluctuating wind record ($1\text{--}8\text{ m s}^{-1}$, AR(1) meander; 1-min steps, a 30-min window). **Wind uncertainty is part of the simulator**: readings are generated under the true wind, but the dataset records only the *measured* wind — truth corrupted by 10° direction and 10% speed error per step, the accuracy class of inexpensive continuous-monitoring anemometry — and no model, at any stage, sees the truth. Because the response is closed-form and linear in rate, the

exact posterior over a 64×64 grid is computable per scenario under known wind (adaptive rate quadrature, float64, normalization error $\sim 10^{-12}$), giving the information floor: 7.0 m at nominal 90% (empirical 0.95), 6.2 m at matched empirical 0.89. A blocking gate requires conformal prediction applied to the exact posterior itself to self-calibrate — a gate that caught two implementation bugs, including a tie-handling error at degenerate scores whose fix we report for reuse. All models share identical data (10k/1k/2k/2k splits, the modal SBI budget [8]), D4 augmentation, and an identical randomized-HPD split-conformal wrapper, calibrated under the same wind condition it is evaluated in. Regions are reported as equal-area circle radii.

B. Diagnosis by elimination (C2)

A DeepSets localizer trained on smoothed one-hot labels is calibrated (coverage 0.89) at 94–98 m — the information for 6–7 m is in the readings, waiting to be extracted. *Is it the estimator head?* A representative flow-based neural posterior estimator on the same encoder — strengthened after adversarial review of our own implementation ($2.5\times$ budget, widened head, full-band conditioning) — reaches 62.8–71.2 m, and we verify its spline head can express 4.4 m regions, so the remaining blur is not head-structural. *Is the classical statistic enough alone?* Conformalizing the matched-filter map with no network returns essentially the entire site: overconfident-wrong, so calibration must be learned. *Is the computation the missing piece?* Localization requires correlating each candidate cell’s space–time signature against all readings — per-cell matched filtering — which pooled set encoders did not recover under the tested architectures and budgets; supplying exactly this computation as input (§III-C) collapses the gap on every encoder. The network cannot compute what the statistic knows; the statistic cannot calibrate what the network learns; the combination is the method.

C. One budget-aware recipe (C3)

For any base localizer (Fig. 2), with g_c the unit-rate response of cell c at the measured wind and $a_c = \sum g_c y$, $b_c = \sum g_c^2$ the linear-Gaussian sufficient statistics: (1) declare the wind-error budget ε (from the anemometer specification), draw $K=8$ plausible true winds from the error model centered on the measurement, and feed four 64×64 channels — the mean matched-filter statistic $\bar{z}_c$, its spread across draws (per-cell wind-fragility of the evidence), mean sensitivity $\overline{\log b_c}$, and the wind-marginalized misfit $\bar{R}_c = yy - 2\hat{q}_c a_c + \hat{q}_c^2 b_c$, $\hat{q}_c = \max(0, a_c/b_c)$ (median-normalized): the residual of the best nonnegative rate fit, averaged over the winds the error model deems plausible. All channels are closed-form linear algebra, milliseconds per scenario, computed from the network’s own inputs. (2) add a 3-layer zero-initialized convolutional head (~ 10 k parameters, 0.2% of the base) summed with the localizer’s logits, so at step 0 the model is the unmodified baseline and every comparison is controlled by construction; (3) train with ordinary smoothed-label cross-entropy — physics enters *only* through the inputs; (4) wrap with split

TABLE I

THE AUDIT AT $\varepsilon=0$: MEDIAN 90% REGION RADIUS VS. THE INFORMATION FLOOR; 2,000 TEST SCENARIOS.

Method	Coverage	Radius (m)
Exact posterior (information floor)	0.89–0.95	6.2–7.0
Deep localizer, labels only	0.89	94–98
Ours ($\varepsilon=0$)	0.89	47.1–47.7

TABLE II

THE GAP IS THE COMPUTATION, NOT THE ESTIMATOR ($\varepsilon=0$; 3 SEEDS PER CELL; COVERAGE 0.88–0.92 THROUGHOUT).

Estimator	Without maps	With maps
DeepSets	94–98	47.1–47.7
GNN	55–58	46.2–47.5
Set Transformer	69–75	46.2–47.3
Flow-NPE (strengthened)	62.8–71.2	57.6–61.2
Classical statistic alone	—	282 (full site)

conformal calibrated under the deployment condition. **At $\varepsilon=0$ the recipe provably reduces to the classical matched filter:** the spread channel is identically zero, and at a single wind the misfit obeys $R_c = yy - \sigma^2 \max(z_c, 0)^2$ — a per-scenario monotone transform of z_c carrying no extra information — leaving exactly the two classical channels ($z_c, \log b_c$). One recipe, continuous in the distance from perfect wind; clean-wind results below are its $\varepsilon=0$ instantiation.

Why not invert the physics directly? On this testbed one can — the exact posterior is our measuring instrument. In deployment, realistic dispersion has no closed form and simulator inversion costs minutes-to-hours per event [10] where monitoring needs seconds — and, as §IV-C shows, plug-in inversion is brittle under wind-measurement error in a way the learned recipe is not.

IV. RESULTS

A. The audit: calibrated is not actionable

Both learned rows pass every calibration check; only the audit exposes that one of them discards an order of magnitude of sharpness (Table I). Two input channels and a 0.2%-parameter head close half the gap: a 50.3% median-radius reduction (bootstrap 95% CI [48.9, 52.0]; 96.4% of scenarios improved), with 51% of individual regions below the 50 m facility scale (percentiles p50/p75/p90: 49/112/170 m — the median case reaches the scale; the upper tail defines the remaining challenge).

B. The missing computation, on every estimator

Without the maps, encoder choice spans 40 m; with them, all nine seed-runs of three architecture families land inside a 1.5 m band (Table II) — the gains are available to whatever architecture a practitioner already has, and encoder design ceases to be the binding constraint. The flow head is a genuine lever without physics yet does not close the gap, and the maps transfer onto it; the classical statistic without a network is overconfident-wrong and pays with full-site regions. Every

TABLE III

MEASURED WIND (10°/10% PER-STEP ERROR; NO METHOD SEES THE TRUTH; CALIBRATION UNDER THE DEPLOYMENT CONDITION; 3 SEEDS FOR LEARNED ROWS). MIDDLE ROWS ABLATE THE BUDGET-AWARE MAPS.

Pipeline	Cov.	Radius (m)	<50 m
Exact plug-in inversion	1.00	282 (full site)	0%
inflated-noise robust variant	1.00	282 (full site)	0%
Network, no physics channels	0.90–0.91	107.9–110.5	0%
Ours, budget ignored	0.91–0.92	82.2–84.2	15–17%
Ours, moment channels only	0.89–0.91	74.7–76.3	20–21%
Ours, full budget-aware	0.90–0.92	70.1–72.9	24–27%

rival explanation fails; the computation was missing, and inputs are where it belongs.

C. Measured wind: the main result

Real anemometers err. Under the 10°/10% budget, plug-in inversion at the measured wind assigns numerically negligible posterior mass to the true source in 30% of scenarios; split conformal preserves its guarantee exactly as theory requires — by returning essentially the entire site (Table III). A tuned global noise-inflation variant does not help: per-step wind error is not absorbable by a scalar noise scale. The learned recipe, by contrast, degrades gracefully, and its accuracy climbs monotonically with how much of the wind-error structure the maps carry: evidence at the reported wind alone, 82–84 m; adding the error-model’s evidence moments (mean and fragility), 74–76 m; adding the marginalized misfit channel, **70–73 m** with a quarter of scenarios below 50 m — same network, same training, same guarantee, one forward pass. The decomposition to remember: 94→47 m is the price of a missing computation, fixed by inputs; 47→71 m is the price of wind uncertainty at this budget, paid at the information level; 108→71 m is what physics inputs are worth when the wind is real. The mechanism is constructive: because the physics — and its declared error model — live in the input representation, clean labels give the network thousands of supervised opportunities to learn the error’s systematic structure and weigh each cell’s evidence accordingly.

V. LIMITATIONS

CH4-T is an instrument, not a deployment: single steady sources (no zero-leak or multi-source scenarios), flat terrain without equipment wakes, known stability class, quasi-static stepping, in-distribution marginal coverage, and a 64×64 grid whose ~7.8 m cells sit near the oracle reference scale (a grid-refinement study is pending). The 6–7 m floor is a property of CH4-T’s generative assumptions, not an atmospheric limit, and it exists only under known wind: a wind-marginalized exact reference is intractable here, so the measured-wind information floor is unknown. The wind-error budget is assumed known and correctly specified (instrument-specification information); sensitivity to a misspecified budget is future work. Half of individual clean-wind cases still exceed 50 m (p90 = 170 m), performance is mildly worse at the highest rates, and architectural claims are scoped to tested encoders and budgets. Field

Figure 2 placeholder (full width): unified pipeline. Readings + measured wind + declared error budget $\varepsilon \rightarrow$ budget-aware physics maps (mean evidence, spread, sensitivity, marginalized misfit; closed form, ms ; at $\varepsilon=0$ reduces to classical $z, \log b$) $\rightarrow$ any localizer + zero-initialized conv head, joined at the logits $\rightarrow$ split-conformal 90% region calibrated under the deployment condition (46–48 m at $\varepsilon=0$; 70–73 m at $10^\circ/10\%$; 50 m facility scale for reference).

Fig. 2. Budget-aware physics-guided calibrated localization (placeholder).

Figure 3 placeholder: measured-wind ladder — dot-range plot of Table III (radius vs. configuration, 3-seed ranges), with the full-site plug-in bar and the 50 m line for scale.

Fig. 3. Accuracy vs. wind-uncertainty structure in the inputs (placeholder).

claims await controlled-release validation — comparisons here are exact regardless, because all models share one verified simulator.

VI. CONCLUSIONS

On a testbed with exact Bayesian ground truth, a calibrated deep localizer delivered a tenth of the sharpness its data support; controlled elimination traced the gap to a single classical computation, and supplying it as input channels brought three architectures into a 1.5 m band at the 50 m facility scale. Made honest about wind — a measurement, never the truth — exact inversion built on the same physics collapsed to the full site, while one budget-aware recipe, provably classical at zero error, held its guarantee at 70–73 m, its accuracy set by how much of the wind marginalization its inputs carry. The constructive message: simulators make exact posteriors available as *instruments* for auditing what learned systems extract; classical signal processing still knows computations networks need not rediscover; and under sensor uncertainty, the place to put physics — including its own error model — is the input. We hope the evidence standard — exact-posterior audits, blocking self-calibration gates, one-change-at-a-time controls — travels furthest.

APPENDIX

APPENDIX: ABLATION AND CONTROL RECORD

DeepSets encoder, identical data and conformal harness, median 90% region radius (m); ^{3s} marks 3-seed ranges.

Reading. Both channels contribute and the matched filter dominates; the sensitivity channel alone ($\log b$, no correlation with the readings) adds little over the baseline, confirming the correlation computation is the operative ingredient. The classical statistic without the network is overconfident-wrong

TABLE IV
COMPONENT ABLATIONS ($\varepsilon=0$), ALL LABEL-TRAINED.

Configuration	Radius (m)	Coverage
Labels only (baseline)	94–98 ^{3s}	0.89
Maps + head (the recipe)	47.1–47.7^{3s}	0.89
z channel only	65.0	0.88
$\log b$ channel only	98.5	0.90
Classical statistic only (no network)	282 (full site)	1.00
Exact-posterior reference	6.2–7.0	0.89–0.95

— the learned calibration is load-bearing. Rate-stratified performance of the recipe (median radius / fraction below 50 m): 10–30 kg h^{−1}: 44.8 m / 54%; 30–100: 44.5 m / 55%; 100–250: 50.3 m / 49%; 250–500: 55.2 m / 41% — the highest emitters localize worst, an operationally relevant tail.

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